

STS

Cuneiform

What is it?

- Cuneiform is one of the oldest forms of writing known. It means "wedge-shaped," because people wrote it using a reed stylus cut to make a wedge-shaped mark on a clay tablet.
- Uruk city - one of the world first true city, located in sumer (southern mesopotamia)

Why do they need it?

- They need it for recording their livestock

How is it used?

- They used clay tablet, reed stylus
- It works where symbols are made
- Early symbols - pictographs; picture of objects

Advancement

- Improved trading where recording became more accurate
- Education and making laws, contracts, and governance
- Legacy - became the foundation of all later writing systems

The Great Ziggurat of Ur & Irrigation and Dikes

Sacred Connection - mountain of god honoring moon god Nanna

Agricultural Necessity

Materials & Structure:

- Sun-ried mud bricks - core
- Fired bricks with bitumen - exterior

Engineering the Sumerian Water System

- Canals - copper and stone tools
- dikes/levees - to contain floodwater
- Shaduf

Agriculture Surplus and Urban Growth

- Year-round farming

AI:

The Great Ziggurat of Ur and the Mesopotamian systems of irrigation and dikes represent the twin pillars of Sumerian society: religious devotion and advanced hydraulic engineering. While the Ziggurat served as the spiritual and administrative heart of the city-state, the irrigation networks provided the agricultural surplus necessary to sustain such monumental construction.

The Great Ziggurat of Ur

Built in the 21st century BCE by King Ur-Nammu and completed by his son Shulgi, the Ziggurat of Ur is one of the best-preserved monuments of ancient Mesopotamia.

- Purpose: It was dedicated to Nanna, the Sumerian moon god and patron deity of Ur. Beyond its religious role, it functioned as an administrative center where agricultural surpluses were collected and redistributed.
- Construction: The structure featured a massive core of mud bricks encased in a protective outer layer of fired bricks set in bitumen (a natural tar) to resist erosion.
- Engineering Marvels: Its design included sophisticated drainage systems and "weeping holes" in the exterior walls to allow moisture to evaporate from the mud-brick core, preventing structural collapse during seasonal rains.

Irrigation and Dikes

The Sumerians transformed the arid plains of southern Mesopotamia into fertile farmland through the world's first large-scale hydraulic engineering projects.

- System Components: They developed complex networks of canals, ditches, and storage basins to divert water from the Tigris and Euphrates rivers to their fields.
- Dikes and Levees: To combat the unpredictable and often violent spring floods, they constructed massive earth embankments (dikes/levees) along the riverbanks. These banks were sometimes reinforced with layers of reeds and bitumen for durability.
- Water Management: They used regulators—including dams and sluice gates—to control water flow. During dry periods, farmers would "poke holes" in the levees or use hoisting devices like the shaduf to irrigate thirsty crops like barley and wheat.
- Social Impact: This mastery of water created a food surplus, which allowed for a division of labor. People who were no longer required for farming became the specialized craftsmen, priests, and government officials who built and managed the Ziggurat.

1. The Great Ziggurat of Ur

At its simplest, a Ziggurat is a massive, terraced step pyramid. Unlike Egyptian pyramids, which were smooth-sided tombs, Ziggurats were solid platforms for temples.

- Ziggurat: Derived from the Akkadian word *zīqquratu*, meaning "to build on a high area." It served as a bridge between heaven and earth.
- Terrace: The flat, raised levels of the structure. The Ziggurat of Ur had three distinct levels.
- Bitumen: A naturally occurring, tar-like substance used as "ancient mortar." It was waterproof, which helped protect the mud bricks from eroding.
- Nanna/Sin: The Moon God. The temple at the very top of the Ziggurat was considered his earthly home.
- Cella: The small, sacred room at the top of the Ziggurat where only priests were allowed to enter to leave offerings for the gods.

2. Irrigation and Dikes

This refers to Hydraulic Engineering, or the science of moving and managing water to grow food in a desert.

- Irrigation: The artificial process of applying water to land to assist in the production of crops. Instead of waiting for rain, Sumerians brought the river to the field.
- Dike (or Dyke): A long wall or embankment built to prevent flooding from the river. Think of it as a "river fence."
- Levee: A natural or man-made slope or wall alongside a river. It is similar to a dike but specifically designed to keep the river within its banks during high water.
- Canal: A man-made waterway. The Sumerians dug massive "trunk canals" (main lines) and smaller "distributary canals" to carry water deep into the desert.
- Silt: Fine sand or clay carried by running water. While it made the soil fertile, it would often clog the canals, requiring constant maintenance and labor.
- Sluice Gate: A sliding gate used to control the flow of water. By lifting or lowering the gate, farmers could "turn on" or "turn off" the water to their fields.

Summary Comparison Table

Term	Category	Primary Function
Ziggurat	Religious	A platform to reach the gods and store surplus grain.
Dike	Engineering	A barrier to stop water from flooding the city.
Canal	Engineering	A path to guide water to the crops.
Bitumen	Material	An adhesive used to waterproof both buildings and boats.

Wheel and Roads

Context and Origin

The wheel

- Mesopotamia (iraq)
- First used as potter's wheel, not for transportation
- Need: overcoming friction

Road

- Earliest paved roads appeared in Ur
- Need: scaling civilization - to overcome obstacle

- “All road leads to rome”

Mechanism and Construction

The wheel

- Solid wooden discs cut from tree trunks, clumped together with wooden struts
- Breakthrough: Axle and spoked wheel

Roads

- statumen(foundation) - lowest layer, stones or rubble
- rudus(base) - smaller stone
- Nucleus (leveling) - layer of sand, concrete
- Summum dorsum(surface) - top layer, resist wear and allow water runoff
- Agger - raised mound - water drainage

Societal Impact

- Warfare - chariot, tank
- Economy - market expanded
- Governance - roads allowed emperor to send messages quickly centralizing power

Legacy

- Modern wheels and roads
- Jnt, wheelchairs

Hanging Gardens of Babylon & Code Hammurabi

Hanging Garden

- Origin - Babylon, Mesopotamia. King Nebuchadnezzar II
- Personal need: ease the homesickness of queen Amytis for her lush homeland
- Societal need - royal wealth
- Materials used: stone & baked bricks, lead sheeting
- It works: archimedes screw
- Just a concept, archeologist haven't found any remnants if it really exist

Code Hammurabi

- King Hammurabi
- Laws
- Clear structure for trade, property rights, and family law
- Materials: Diorite stele - black stone monument
- Laws are carved in akkadian cuneiform and placed in public places (temples)
- Governance: established the “Rule of Law”, defined social classes and standardize trade and labor

Papyrus & Clepsydra

Papyrus

- Origin: egypt
- Made from pith of cyperus papyrus plant
- Writing surface
- Revolutioned record keeping
- Needed: used to record, trade transactions
- Materials: papyrus plant, water, stone or wooden tools for pressing
- Enable education and communication egyptian empire
- Modern wooden “paper” comes from papyrus

Clepsydra

- Water clock
- Ancient timekeeping device that measures time by the regulation of water into or out of a vessel
- Indicate the passage of hours
- It is used for measuring time at night because sundials does not work indoor
- Used for: religious rituals, work schedules
- Materials: stone water, small holes, marked line inside the container
- How it works:
- Allowed people measure time day and time
- Act like a timer, inaalang kung gaano katagal ng oras ang lumipas

Hieroglyphics & Calendar

Hieroglyphics

- Writing system of egypt
- Nile river valley (north africa)
- Enabled written communication
- Written in stones and papyrus
- Types of symbols: phonograms (representing sounds), logograms (representing whole words or concepts), and determinatives (silent signs providing context)

Calendar

- Calendar helps predict flooding and farming seasons
- 3 seasons flooding planting harvest, Akhet (Flood/Inundation, mid-July–Nov) covered fields in nutrient-rich silt, Peret (Emergence/Planting, Nov–March) was when crops were sown, and Shemu (Harvest, March–mid-July)

Alarm Clock and Water Mill

Water Mill

- Greece
- How human interacted with nature to produce energy
- Need: ground grain into flour
- Works: water power -> gearing up
- Materials: local wood
- Hydraulics

Alarm Clock

- Starts with plato
- Greece
- Added alarm feature to clepsydra
- Needed a standardized time in law
- First automated notification system

Roman Architecture & Roman Numerals

Roman Architecture

- Romans are known for wars, for being a spartan
- Coliseum
- Well known designs - arcs, they believed that it balances the weight

Roman Numerals

- Number system - sticks, dangkal
- To have a standardized number system

Silk & Tea Production

Silk

- "luxury"
- Aka sericulture
- Silkworms
- Neolithic china
- Credited to lady of hsi leizu
- Textile for clothing
- Bombyx mandarina - picky eaters and eats white mulberry leaves
- Process: cultivation, stifling and boiling, reeling, degumming and weaving
- Birth of silk road

Tea

- Medicinal necessity - detoxify, improve digestion
- Camella Sinensis Leaf

- Process: steaming, compression, preparation
- Tea as currency because it is being exported

Great Wall of China & Gunpowder

Great Wall of China

- Defensive wall system
- Keeping “barbarian” customs
- Need: nomadic invasions(xiongnu, mongols), protect farmland, trade routes, political power
- Materials: rammed earth, stone, bricks, wood, lime, gravel
- Built by soldiers, peasants, prisoners
- Early example of large-scale state engineering

Gunpowder

- Chemical explosive made from saltpeter, charcoal, sulfur
- Need: medicine, spiritual rituals, military purposes
- Tang dynasty chinese - birth of gunpowder
- When heated it rapidly produces gas and heat
- Used in fireworks, bomb
- Reduce the dominance of knights and castles
- Enabled dev of guns and weapons

Heavy Plough & Hourglass

02/18/2026

Astrolabe & Compass

Astrolabe

- Early scientific instrument
- Use in observing the position of sun and stars
- Ancient greece, used to study stars and planets
- First recorded use - Hipparchus
- Made from brass and iron
- Rete - heavens and acts as celestial map

Compass

- Shows the cardinal direction
- The needle is always pointed at the north
- Made from magnetized metal typically iron

Eyeglass & Spinning Wheel

Eyeglass

-

Spinning Wheel

-

Mechanical Clock & War Weapons

Mechanical Clock

- Back then, people rely on sundials (sun) & clepsydra (water)
- Why is a clock needed? Need and accurate hr for their prayer cycle (7-8)
- purely mechanical machine that uses weight and gears, replaced the flow of water with the rhythm of machinery
- Verge and foliot escapement - the heart of the clock
- Impact: created the concept of an "hour", time is money - gives birth to capitalism, monastery -> marketplace

War Weapons

- High medieval
- Simple leather jerkins -> chainmail (interlocking rings) -> Plate armor
- Chainmail - vulnerable to blunt force and piercing points
- Impact: social equalizer, professional science, rise of nation-state

Windmills & Printing Press

Printing press

Pre- Innovation era

- Books were handwritten by monks
- Expensive and rare
- Only the elite and church had access to knowledge, knowledge = power
- Printing press - a machine that allows mass production of written text

Windmills

- Appeared in Persia
- Machine that converts wind energy into mechanical power
- Grinding grains, pumping water, later generating electricity
- Reduce human and animal labor
- Enable land reclamation (process of creating new land)
- Mahatao, Batanes - first commercial wind-diesel hybrid power plant (2004)
- Ilocos and Rizal - first major installation in PH (2005)
- Basco, Batanes - first magnus vertical-axis wind turbine in PH (2021)

Inventions of Leonardo da Vinci & Anatomy

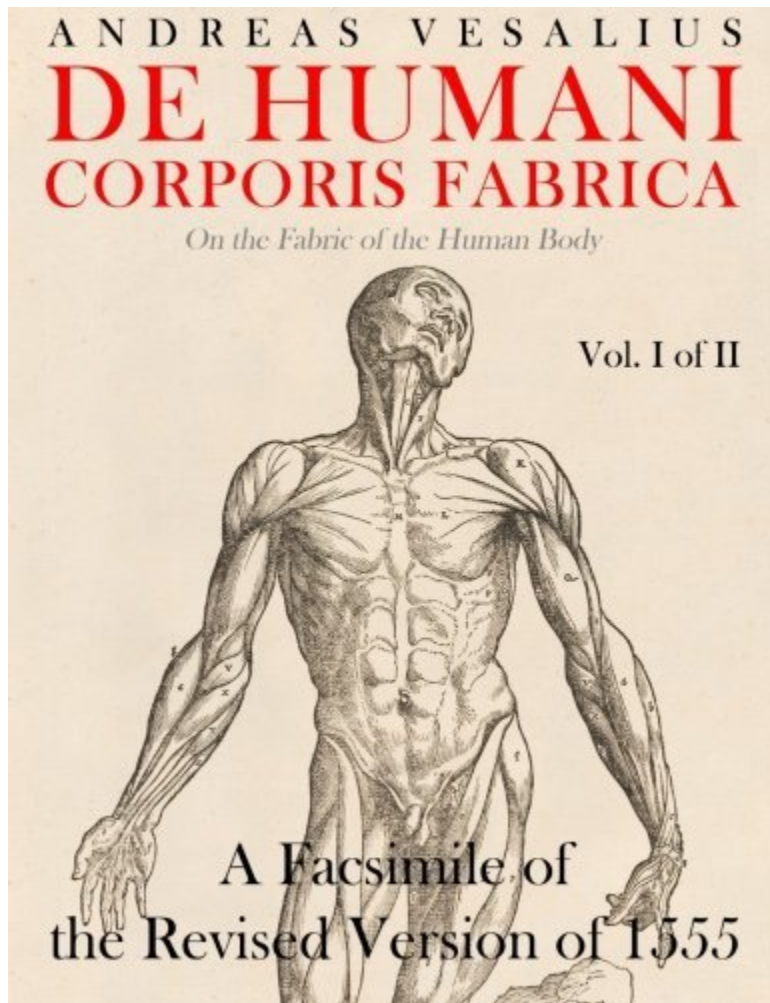
Leonardo da Vinci - seen as the model of the 'Renaissance Man'

Inventions:

- Flying machine (Ornithopter) - inspired by bird wings
- Armored tank - circular design with mounted cannons
- Helicopter (Aerial Screw) - vertical flight
- Military Machine - giant crossbow, machine gun-like weapon
- Hydraulic Systems - canals, water pumps, bridges

- Introduced scientific engineering based on observation
- Military design
- Technology could not support his ideas
- Used in helicopter, tanks, robotics

Anatomy - book of Andreas Vesalius



Book 1 - bones & cartilages

Book 2 - ligaments & muscles

Book 3 - veins & arteries

Book 4 - the nerves

Book 5 - digestive & reproductive

Book 6 - Heart & Associated Organs

Book 7 - The brain & sense organs

- Dissection was rare and forbidden by the church
- Many anatomical drawing are inaccurate
- Medicine relied on theory than observation

What did he do?

- He performed dissection of human cadavers
- Created 240 detailed anatomical drawings
- Studied: muscles, skeleton, heart, etc.
- He is known as the Pioneer of modern science
- Observe -> Record -> Analyze
- He blended art, science, and society

- He is one of the people who harness energy from the sun, fire

Heliocentric Model & Astronomical Observation

Heliocentric Model

- Before people believe that earth is the center of the universe - **geocentric** (Earth-centered)
- All planets and the sun revolve around Earth
- Proposed by Nicolaus Copernicus (1543)
- Sun is at the center
- Planets revolve around the sun
- Earth rotates on its axis
- Philosophy -> scientific evidence
- It is the theory that the sun is at the center of the solar system
- Introduced to explain the planetary motion more accurately than the geocentric model

Astronomical Observation

- telescope

Microscope & Telescope

Telescope

- Making of lenses during 17 century lead by spectacle-making hub
- Looker
- Hans Lippershey - first invention of telescope by diff combination of lenses
- Janssen Family - known for making glasses, Zacharias and Hans -> compound microscope
- Lens - refraction of light
- Convex objective lens

Microscope

-

Microscopy & Inventions of Galileo Galilei

Galileo

- Four largest moon of jupiter
- Motion, astronomy, and strength of materials

Microscopy

- the technical field of using microscopes to view, magnify, and analyze objects too small for the naked eye, such as cells and materials
- Types:
 - Light - most common form, uses visible light. Used in school
 - Elentron - use to see very small specimen
 - SPM (Scanning Probe Microscope) - use a scanner to zoom in things

Inventions of Galileo Galilei

- improved refracting telescope - increased by 20x magnification
- the thermoscope (early thermometer) - didn't have numbers yet
- the military compass aka sector
- hydrostatic balance - measuring density
- early microscopes (occholino), while also developing foundational principles for pendulums and, conceptually, clocks

Laws of Motion & First Steam Engine

Laws of Motion

- Proposed by Isaac Newton - gravity
- Change in position of an object
- Explain how things move and react to push and pull
- Improves transpo
- First Law (Law of Inertia): An object at rest stays at rest, and an object in motion stays in motion with a constant velocity unless acted upon by an external, unbalanced force.
- 2nd Law (law of acceleration) - $F = ma$; Mas bumibigat ang bagay, mas malaking force ang need
- 3rd law (action & reaction) - For every action force, there is an equal and opposite reaction force; 'pag sinampal ko si eyang magre-react si nikol (YES)
- Newton vs aristotle - A: common sense, N: math

First Steam Engine

- Mechanical device to create power from boiling water

- Was invented to pump water out of flooded mine
- Sparked the start of industrial revolution
- Problem: Mas malalim ang minimina, may tubig, bumabaha. Solution: Water pump

Lightning Rod & Textile Industry Inventions

Lightning Rod

- a pointed metal rod, usually copper or aluminum, mounted on top of a structure to protect it from damage
- Placed on top of structures to direct lightning into the ground
- Invented by Benjamin Franklin
- To prove that lightning is a form of electricity

Textile Industry Inventions

- Pre-innovation: Hand-spun and woven
- Spinning Jenny - James Hargreaves - spun multiple threads at once
- Water Frame - Richard Arkwright - water power to spin
- Spinning mule - Samuel Crompton, hybrid, crossing the technology of the spinning jenny and the water frame
- Power Loom - Edmund Cartwright, automated weaving process
- Why Invent? Padaliin ang woving
- Manual weaving -> using machinery for fast weaving

Smallpox Vaccine & Development of the Steam Engine - wannabee tutubi mwa

Smallpox Vaccine

- Dr. Edward Jenner
- Smallpox - virus caused by variola virus (bad virus), contagious
- $\frac{1}{3}$ died
- Variolation - deliberately infecting a healthy person with material from a patient with a mild case
- Uses vaccinia virus - good virus
- Resistance - Turn people into cows HAHAH

Development of the Steam Engine

- James Watt
- Added spinning wheel, and condenser, valves / improve
- Human vs. Machine - Displacement and loss of their livelihood

Atomic Theory & Electromagnetism

Atomic Theory

- All things are made up of atoms proposed by Democritus
- John Dalton - First modern atomic theory
- J.J Thompson - Discovered electron
- Atomic Theory - atoms, molecule, chemical reaction

Electromagnetism

- Michael Faraday o electromagnetic induction
- James Clark Maxwell - electricity and magnetism through equation

Telegraph & Photography

Telegraph

- Samuel Morse and Morse Code - Developed and assigned dots and dash (miscommunication)
- Cooke and Wheatstone - developed needle system for railroad signing
- an apparatus for communication at a distance by coded signals, morse code
- Sparked the start of long distance communication

Photography

- Joseph Nicéphore Niépce pioneer of photography
- Heliography -
- Kodak Camera, Roll Film,

Theory of Evolution & Germ Theory

Theory of Evolution

- Before, people believed that all living things were thought of unchanging
- Charles Darwin - developed the theory
- species change over time through descent with modification, originating from common ancestors
- Heliocentric model image placeholder (shuttershock)
- Conflict: Religious groups believed contradicting creation stories

Germ Theory

- states that specific microscopic organisms (pathogens like bacteria, viruses, fungi, or protists) are the cause of specific infectious diseases
- People believed that diseases were caused by bad air (miasma)
- Louis Pasteur - Developed the theory

- Resistance - Doctors believed miasma theory and refused invisible microorganisms
- Impact – Improved sanitation and hygiene

Telephone & Electric Light Bulb

Telephone

- Patented by Alexander Graham Bell
- “MR. WATSON, COME HERE I WANT TO SEE YOU”
- Elisha Gray - other inventor but not patented
- Bell Telephone Company - Bell created telecommunication industry

Electric Light Bulb

- Invented by Thomas Edison, 1879
- Why invent? Fire is a hazard; candles and flames > fire
- Industrial Revolution

Aircraft & Television - saji boi bibiboi

Aircraft

- Vehicle designed to fly
- Wright brothers, orville & wilbur
- Used to send goods, used for trading

Television

- Electronic device that receives and decodes visual and audio signals
- Invented by John Logie Baird & Philo Farnsworth
- They rely on newspaper before television

Penicillin & Nuclear Fission

Penicillin

- An antibiotic that kills bacteria
- Alexander Fleming - Develop penicillin
- Later developed by Howard Florey
- Why invent? During WW2, soldiers were killed despite minor wounds
- Saved millions using modern medicine
- How does it work? Breaks cell wall causes bacteria to die

Nuclear Fission

- Process of splitting atoms
- Used for electricity
- Otto Hahn, Otto Frisch - Discovered nuclear fission
- Why invent? Understand atomic energy, for powerful weapon, and electricity
- Negative - use to create nuclear weapon

- Fission vs Fusion: Fission splits atoms; Fusion combines them.

DNA Structure & Space Exploration

DNA Structure

- Biological blueprint
- Deoxyribonucleic acid
- The body's guide and instruction book
- Double helix
- James Watson and Francis Crick
- Looks like twisted ladder
- Adenine, Thymine, Guanine Cytosine - G:C, T:A

Space Exploration

- Study of outer space using satellites and astronauts
- 1957, year started
- Neil Armstrong - first man to walk on moon
- Escape velocity - must move very fast to escape earth's gravity
-

Internet & Personal Computer

Internet/Interconnected networks

- Arpanet - first network, 1969
- TCP/IP allowed diff networks to connect together, 1980
- Berners lee - created WWW, 1991

Personal Computer

- 1970 - early small computers were made for hobbyist
- 1977 - apple inc. released user-friendly home computers
- 1981 - IBM launched the IBM PC, setting the standard
- 1980s - 1990s - microsoft made windows, making computers easier to use
- How it works: input from keyboard and mouse -> Processing -> storage

Human Genome Project & Smartphones

Human Genome Project anliit wala ako mabasa

Google ai: The Human Genome Project (HGP) primarily used a hierarchical shotgun sequencing approach (clone-by-clone) to map and sequence the 3 billion base pairs of the human genome, completed in 2003. This method broke DNA into large chunks, mapped them to chromosomes, and then shotgun-sequenced each piece for high accuracy.

- Shotgun sequencing - technique used

- Francis Collins (Project Leader) & J. Craig Venter (Private Company Scientist) - key inventors
- Process: Break DNA into pieces -> use super-fast DNA reading machine -> reassemble with powerful computers
- Importance: genetic blueprint, medical diagnostics, human history

Smartphone

- Brick Phone (1983) - 30mins talk time, 10hrs charging time
- Martin Cooper - inventor of first mobile call, 1973
- Frank J. Canova - designer of smartphone
- Steve Jobs - iPhone Launch, 2007

CRISPR - CAS9 & Electric Vehicle - eyang tomboy

CRISPR - CAS9

- Dna - blueprint of humans
- Human genome project - mapping out all of human genes
- Designer baby - modifying the genes of humans, choosing the perfect genes for your baby
- CRISPR-CAS9 - used to repair the disease that are inherited
- Pagiging bading - communicable disease

Electric Vehicle

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Renewable Energy & Artificial Intelligence

Renewable Energy

- Solar Power (sun), Wind Power (wind), hydropower (water), geothermal (ilalim ng lupa)
- 1970 Oil Crisis - prices quadrupled overnight in western nations which leads them to use renewable energies
- Benefits: cleaner air, new jobs, helping farmers, power for remote areas

Artificial Intelligence

- When AI do tasks that need human intelligence
- Why create? Understand intelligence, handle huge amount of data
- Alan Turing - "Can machines think?"
- When philosophy meets technology; philosophy + technology = AI
- Benefits: Healthcare, VA, Navigation, Shopping, Photos, Translation, Security

Space Exploration & Neuroscience

Space Exploration

- Discoveries: Reusable rockets, can land and be reused: Falcon 9
- International Collaboration: lab in orbit
- Mars exploration: flight in another planet
- Satellite
- NASA - National Aeronautics and Space Administration
- Diwata 1 - first satellite of the PH
- Iwata 2 - image acquisition

Neuroscience

- Study of the nervous system, especially the brain
- Biology + psychology
- Santiago Ramon y Cajal - identified neuron structure
- Eric Kandel - explained memory processes
- Neurons - nerve cells
- Cerebrum - thinking & memory, Cerebellum - muscle movements, brainstem - breathing
- Modern tools: MRI, fMRI, EEG, CT Scan

Banana Ketchup & Medical Incubator

Banana Ketchup

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Medical Incubator

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Erythromycin & Mole Remover

Erythromycin

-

Mole Remover

-

Electronic Jeepney & Mosquito Ovicidal Trap

Electronic Jeepney

-

Mosquito Ovicidal Trap

-

SALT Lamp & Salamander Amphibious Tricycle

SALT Lamp

-

Salamander Amphibious Tricycle

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Copernican Revolution & Darwinian Revolution

Copernican Revolution

-

Darwinian Revolution

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Freudian Revolution & Science and Technology in the Philippines During Pre-Colonial Period

Freudian Revolution

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Science and Technology in the Philippines During Pre-Colonial Period

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Science and Technology in the Philippines During Colonial Period & Post Colonial Period

Science and Technology in the Philippines During Colonial Period

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Post Colonial Period

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Angel Alcala & Ramon Barba

Angel Alcala

-

Ramon Barba

-

Tetchi Cruz - Capellan & Edgardo Gomez

Tetchi Cruz - Capellan

-

Edgardo Gomez

-

Aisa Mijeno & Fabian Dayrit

Aisa Mijeno

-

Fabian Dayrit

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Digital Design

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Number Systems & Radix Theory

02/26/2026

Data Integrity & Encoding